CONTROL NARRATIVE DOCUMENT

Pump Station PS-01 — Rev C

Document Number: CND-PS01-003

Revision: C

Date: 29 October 2026

1. GENERAL DESCRIPTION

Pump Station PS-01 transfers process fluid from suction vessel T-100. Two centrifugal pumps operate duty/standby. Flow regulated by FV-301.

1.1 P&ID Schematic

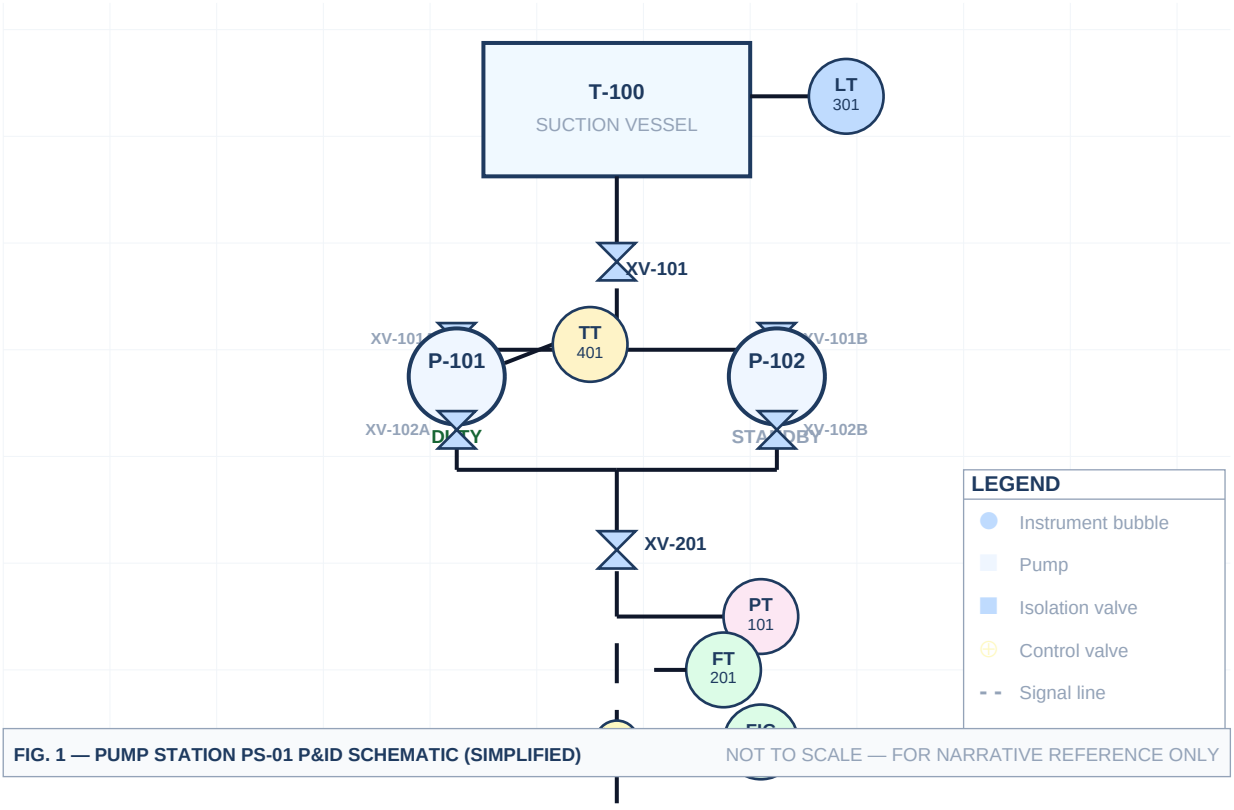

FIG. 1 — PUMP STATION PS-01 P&ID SCHEMATIC (SIMPLIFIED) NOT TO SCALE — FOR NARRATIVE REFERENCE ONLY

2. EQUIPMENT LIST

Tag	Description	Type	Range	Normal State
P-101	Duty pump	Centrifugal pump	0–150 m³/h	Running
P-102	Standby pump	Centrifugal pump	0–150 m³/h	Stopped
XV-201	Main discharge valve	Ball valve	DN200, PN16	Open
FV-301	Flow control valve	Pneumatic globe valve	Cv 120	Auto
PT-101	Discharge pressure	Pressure transmitter	0–300 PSI	Active
FT-201	Process flow	Magnetic flow meter	0–200 L/min	Active
LT-301	Suction vessel level	Radar level gauge	0–5 m	Active

3. OPERATING MODES

3.1 STOPPED

Both pumps stopped. XV-201 closed.

3.2 STARTING

Startup sequence in progress.

3.3 RUNNING

P-101 running, FIC-201 controlling to 80 L/min.

3.4 STOPPING

Shutdown sequence executing.

3.5 EMERGENCY SHUTDOWN

All pumps stopped immediately. Manual reset required.

4. SETPOINTS TABLE

Tag	Parameter	Value	Units	Ref
PT-101	High High Trip	145	PSI	p.3
PT-101	High Alarm	130	PSI	p.3
FT-201	Low Flow Trip	10	L/min	p.3
FT-201	Low Alarm	25	L/min	p.3
LT-301	Low Low Trip	15	%	p.3
TT-401	High High Trip	85	°C	p.3

5. INTERLOCKS

Interlock IL-101 activates when PT-101 discharge pressure exceeds 150 PSI for more than 3 seconds. Action: stop pump P-101 AND P-102 AND close XV-201. Reset condition: PT-101 below 120 PSI AND manual reset by operator.

Interlock IL-201 activates when FT-201 flow rate falls below 20 L/min for more than 5 seconds. Action: stop P-101 AND auto-start P-102. Reset: automatic after P-102 restores flow.

6. ALARMS

Tag	Type	Setpoint	Units	Priority	Operator Response
PT-101-HH	HH	150	PSI	Priority 1	Stop all pumps. Close XV-201. Investigate.
PT-101-H	H	130	PSI	Priority 2	Reduce FIC-201 setpoint. Monitor.
FT-201-L	L	20	L/min	Priority 2	Check FV-301 and pump status.
LT-301-L	L	25	%	Priority 2	Contact feed ops.

7. STARTUP SEQUENCE

Step	Action	Detail
Step 1	Check permissives	Verify LT-301 > 25%, no active ESD.
Step 2	Open XV-101	Open suction valve. Confirm open within 15 s.
Step 3	Start P-101	Start duty pump. Confirm running within 10 s.
Step 4	Open XV-201	Open the gate valve XV-201 to allow flow.
Step 5	Enable FIC-201	Enable flow controller in AUTO at 80 L/min.

8. NORMAL SHUTDOWN SEQUENCE

Step	Action	Detail
Step 1	Ramp down flow	Reduce FIC-201 to 20 L/min.
Step 2	Close FV-301	Switch to MANUAL, close FV-301.
Step 3	Stop P-101	Stop P-101 BEFORE closing XV-201.
Step 4	Close XV-201	Close XV-201 after P-101 has stopped.
Step 5	Close XV-101	Close XV-101. Verify closed within 15 s.

9. EMERGENCY SHUTDOWN (ESD)

Step	Action	Detail
ESD-1	Stop all pumps	De-energise P-101 AND P-102 simultaneously.
ESD-2	Close XV-201	Hardwired signal to XV-201. Closes immediately.
ESD-3	Close XV-101	Close suction isolation.
ESD-4	Latch ESD	Set ESD latch. Manual reset required.